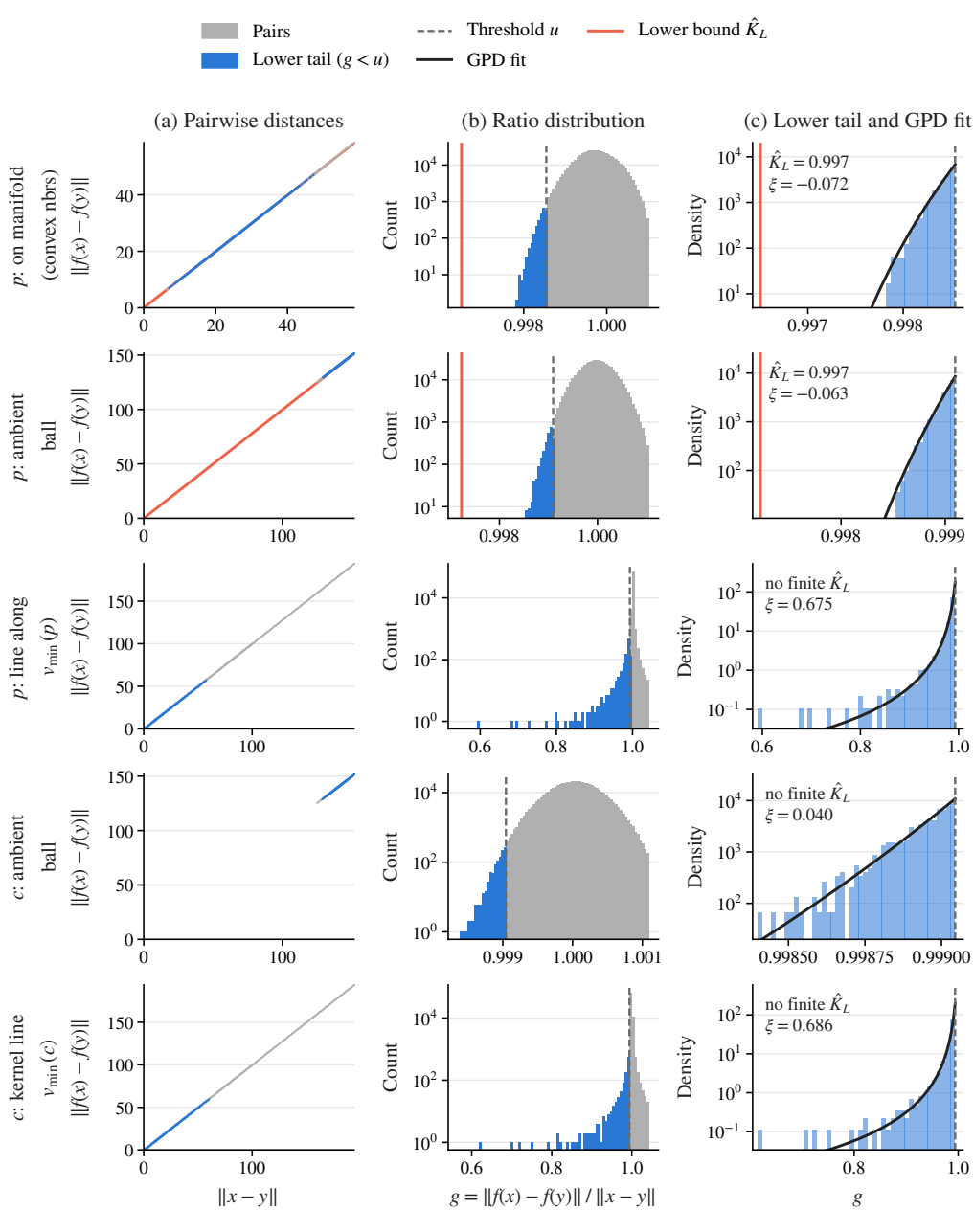

Pairs
Lower tail ($g < u$)
Threshold $u$
GPD fit
Lower bound $\hat{K}_L$
(a) Pairwise distances
(b) Ratio distribution
(c) Lower tail and GPD fit
$p$: on manifold (convex nbrs)
$p$: ambient ball
$p$: line along $v_{\min}(p)$
$c$: ambient ball
$c$: kernel line $v_{\min}(c)$
$\|f(x) - f(y)\|$
Count
Density
$\hat{K}_L = 0.997$
$\xi = -0.072$
$\hat{K}_L = 0.997$
$\xi = -0.063$
no finite $\hat{K}_L$
$\xi = 0.675$
no finite $\hat{K}_L$
$\xi = 0.040$
no finite $\hat{K}_L$
$\xi = 0.686$
$\|x - y\|$
$g = \|f(x) - f(y)\| / \|x - y\|$
$g$